02 – Feature engineering

Phase 4: the feature pipeline is fitted on the training rows only and applied to the validation rows. This notebook re-fits it and checks that it gives exactly the saved features.

In [1]:
```
# Setup: work from the project root so the project code and configs are found
import os, sys, json, platform, warnings
from pathlib import Path
ROOT = Path.cwd().parent if Path.cwd().name == "notebooks" else Path.cwd()
os.chdir(ROOT); sys.path.insert(0, str(ROOT))
warnings.filterwarnings("ignore")
import numpy as np, pandas as pd
from IPython.display import Image, Markdown, display
pd.set_option("display.precision", 3); pd.set_option("display.width", 160);
pd.set_option("display.max_columns", 20)
from src.utils.config import load_config, resolve_path
REPORTS = resolve_path(load_config("project")["paths"]["reports"])
FIG = resolve_path(load_config("project")["paths"]["figures"])
print("Project folder:", ROOT)
print("Python", platform.python_version(), "| numpy", np.__version__, "|
pandas", pd.__version__)
```

```
Project folder: C:\AL_outcome_predictor
Python 3.12.0 | numpy 2.5.4 | pandas 3.0.6
```

1. Re-fit the pipeline on the training rows

In [2]:
```
from src.data.split import load_split_rows
from src.features.build import FeaturePipeline
train, validation = load_split_rows("train"), load_split_rows("validation")
pipe = FeaturePipeline(load_config("features")).fit(train)
x_train, x_val = pipe.transform(train), pipe.transform(validation)
print("Training rows:", x_train.shape, "| validation rows:", x_val.shape)
```

```
Training rows: (709, 98) | validation rows: (152, 98)
```

2. Same result as the saved features (used for training)

In [3]:
```
processed = resolve_path(load_config("project")["paths"]["processed"])
for name, fresh in (("train", x_train), ("validation", x_val)):
    saved = pd.read_csv(processed / f"features_{name}.csv")[fresh.columns]
    diff = np.abs(saved.to_numpy(float) - fresh.to_numpy(float)).max()
    print(f"{name}: largest difference from the saved features = {diff:.2e}")
```

```
train: largest difference from the saved features = 4.44e-16
validation: largest difference from the saved features = 4.44e-16
```

3. Feature sets compared in Phase 6, and the 34 columns finally used

In [4]:
```
sets = json.loads((REPORTS / "phase04" / "feature_sets.json").read_text("utf-8"))
print("Feature set sizes:", sets["sizes"])
final = json.loads((REPORTS / "phase06" / "final_columns.json").read_text("utf-8"))
print(len(final), "columns used by the final models:")
pd.DataFrame({"column": final})
```

```
Feature set sizes: {'full': 97, 'composites': 85, 'filtered': 34}
34 columns used by the final models:
```

Out[4]:

	column
0	ol_first_language
1	ol_maths
2	ol_history
3	ol_science
4	ol_cat1
5	ol_cat2
6	ol_cat3
7	class_rank_freq
8	g11_self_study_hours
9	career_goal_clear
10	g6_11_perf_rating
11	er1
12	er3
13	er4
14	ol_core_mean
15	ol_sci_maths_mean
16	self_efficacy_index
17	effort_regulation_index
18	interest_index
19	ol_n_A
20	motivation_index
21	goal_clarity_index
22	school_performance_index
23	stream_influence_personal_interest
24	stream_influence_teachers
25	ol_school_type_International school
26	ol_school_type_National school
27	ol_school_type_Other
28	ol_school_type_Private or semi-government school
29	ol_school_type_Provincial school
30	al_subject_combination_Biology + Chemistry + A...
31	al_subject_combination_Biology + Chemistry + P...
32	al_subject_combination_Chemistry + Physics + C...
33	al_subject_combination_Physics + Combined Math...

4. How each feature is made (catalogue, first rows)

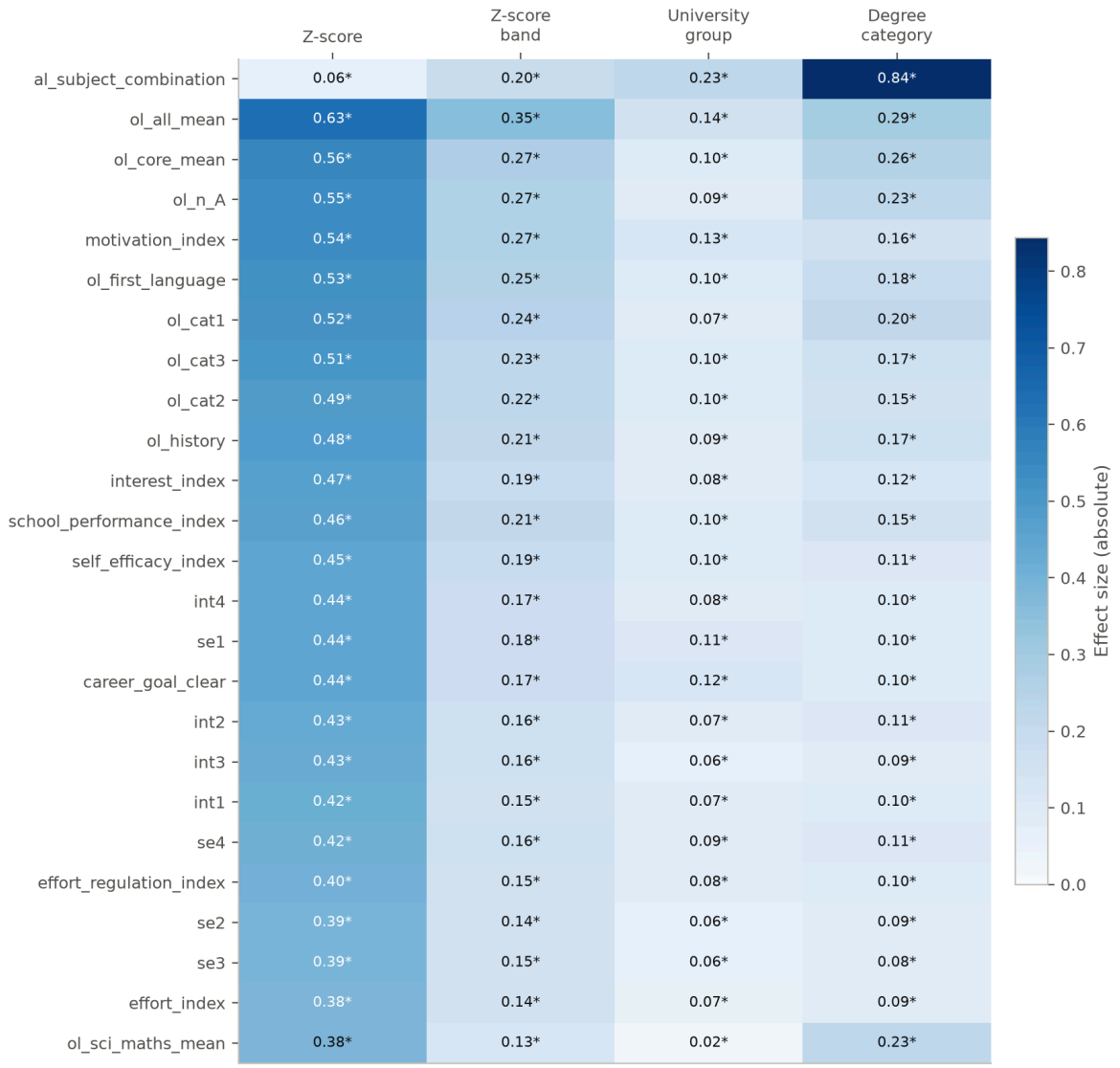
In [5]:
```
cat = pd.read_csv(REPORTS / "phase04" / "feature_catalogue.csv")
cat[cat["feature"].isin(final)][["feature", "type", "formula"]].head(15)
```

Out[5]:

	feature	type	formula
1	ol_first_language	ordinal (standardised)	ordinal code, median-imputed, standardised
3	ol_maths	ordinal (standardised)	ordinal code, median-imputed, standardised
4	ol_history	ordinal (standardised)	ordinal code, median-imputed, standardised
5	ol_science	ordinal (standardised)	ordinal code, median-imputed, standardised
6	ol_cat1	ordinal (standardised)	ordinal code, median-imputed, standardised
7	ol_cat2	ordinal (standardised)	ordinal code, median-imputed, standardised
8	ol_cat3	ordinal (standardised)	ordinal code, median-imputed, standardised
9	class_rank_freq	ordinal (standardised)	ordinal code, median-imputed, standardised
10	g11_self_study_hours	ordinal (standardised)	ordinal code, median-imputed, standardised
12	career_goal_clear	ordinal (standardised)	ordinal code, median-imputed, standardised
13	g6_11_perf_rating	ordinal (standardised)	ordinal code, median-imputed, standardised
19	er1	ordinal (standardised)	ordinal code, median-imputed, standardised
21	er3	ordinal (standardised)	ordinal code, median-imputed, standardised
22	er4	ordinal (standardised)	ordinal code, median-imputed, standardised
27	ol_core_mean	composite (standardised)	mean of the six compulsory O/L grades (0-4)

In [6]:
```
display(Image(str(FIG / "phase04_relevance.png"), width=760))
```

Strongest training-set associations (* = significant after FDR correction)



Measures: Spearman / point-biserial (Z-score); epsilon-squared or Cramer's V (class targets). Association, not causation.